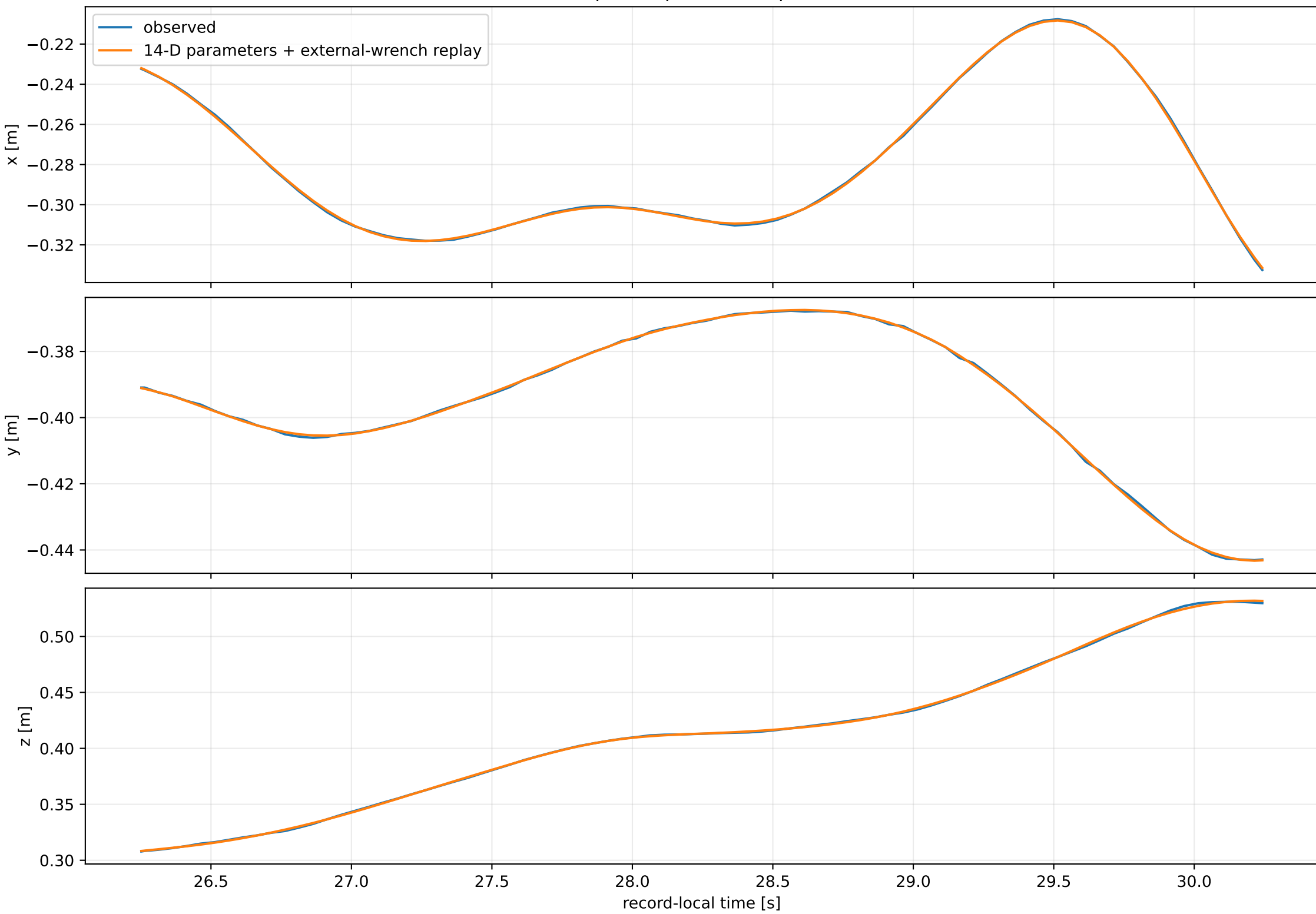
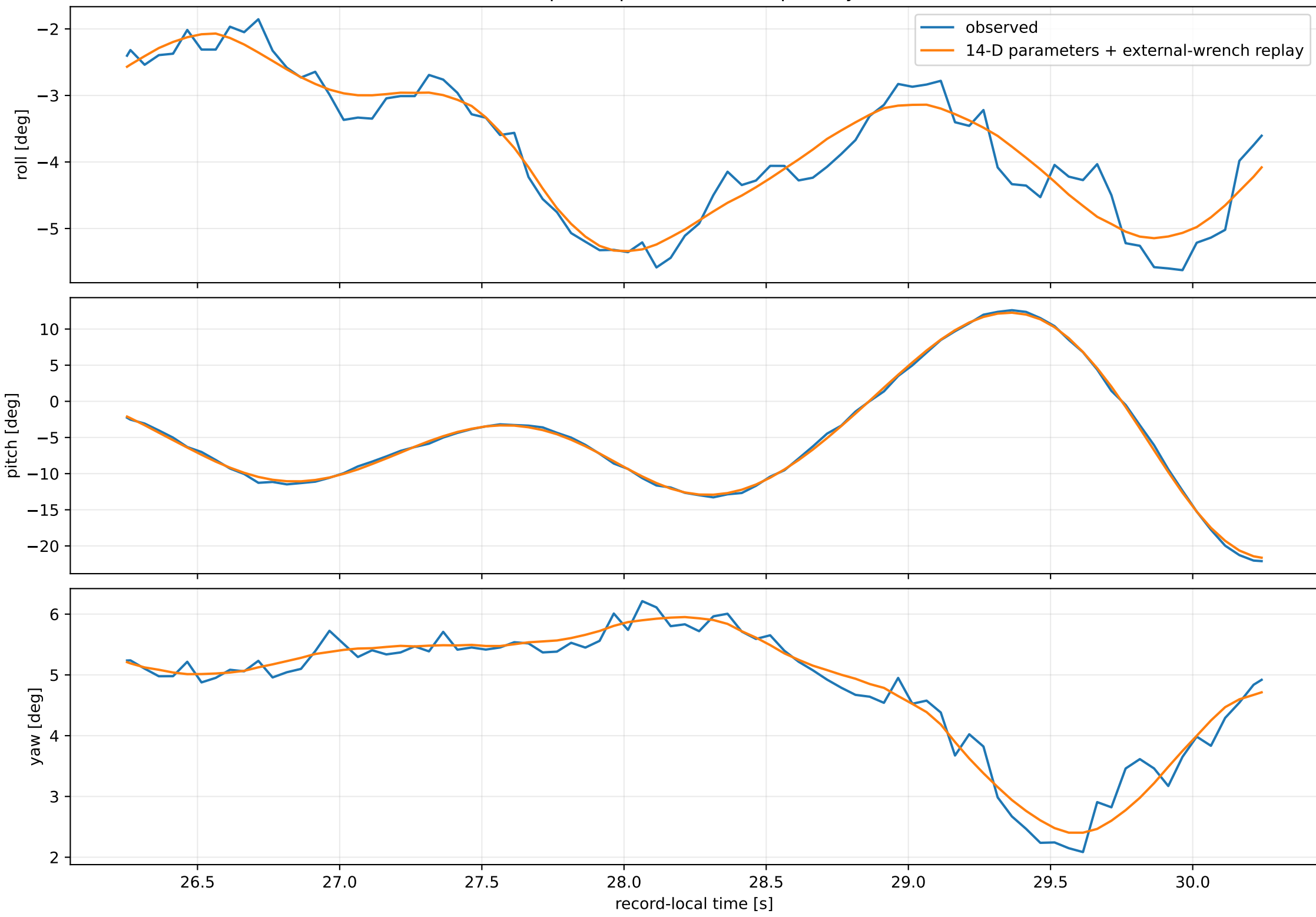


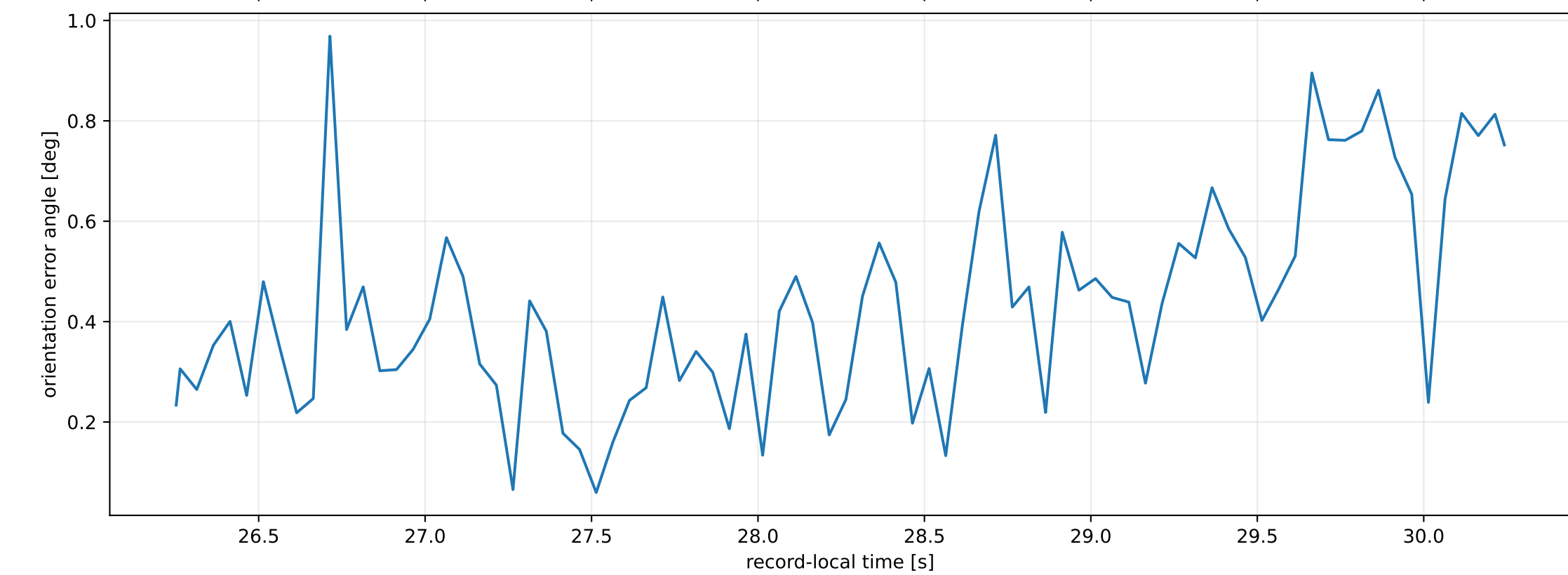
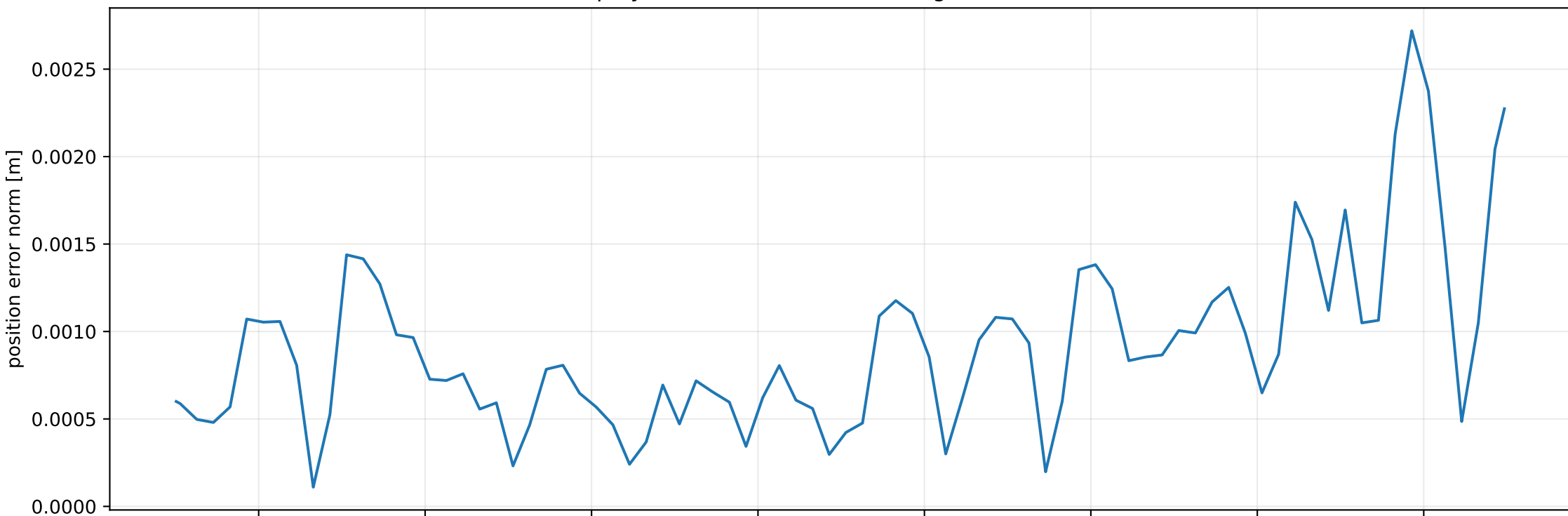
Fine pose reproduction: position



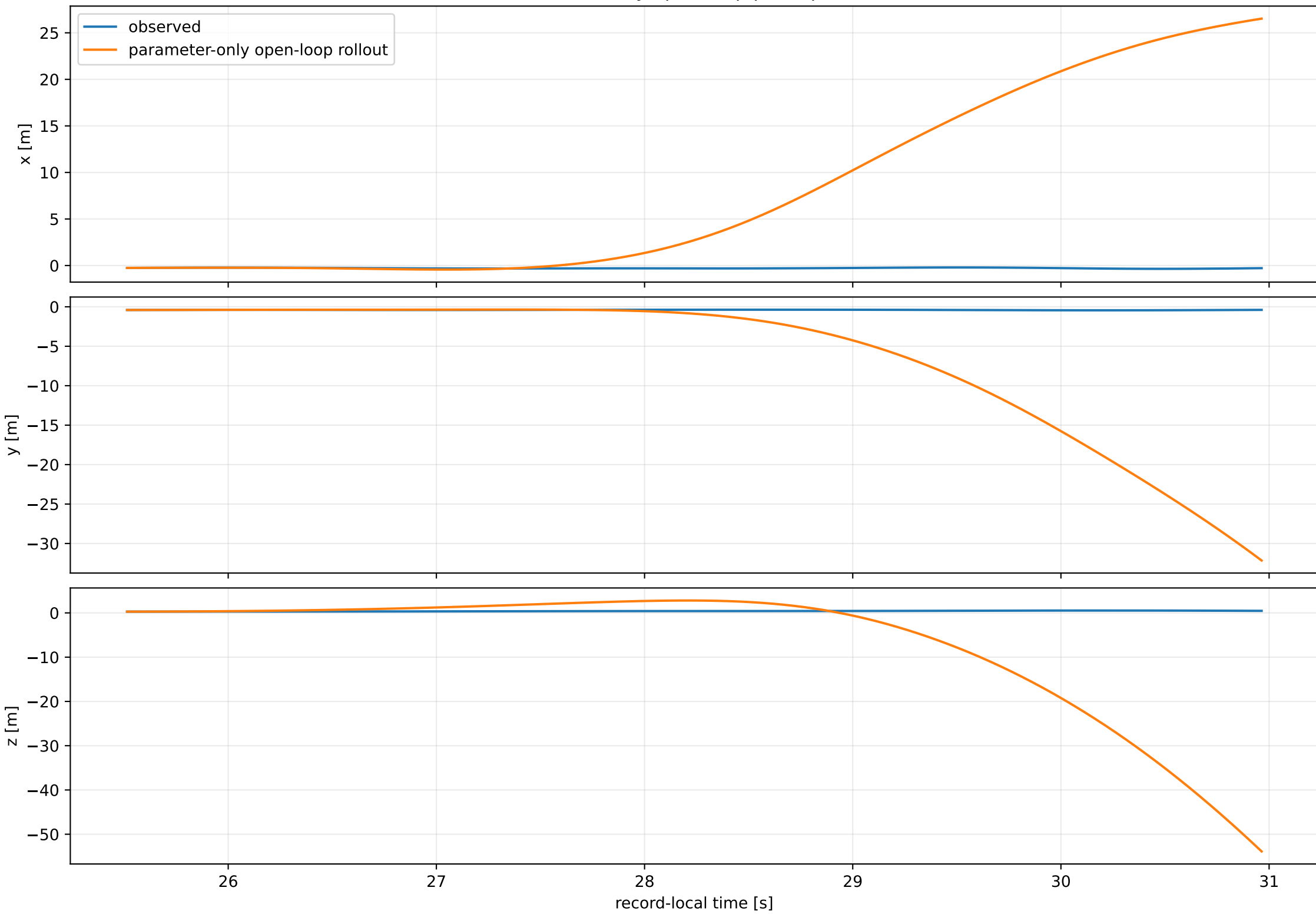
Fine pose reproduction: roll / pitch / yaw



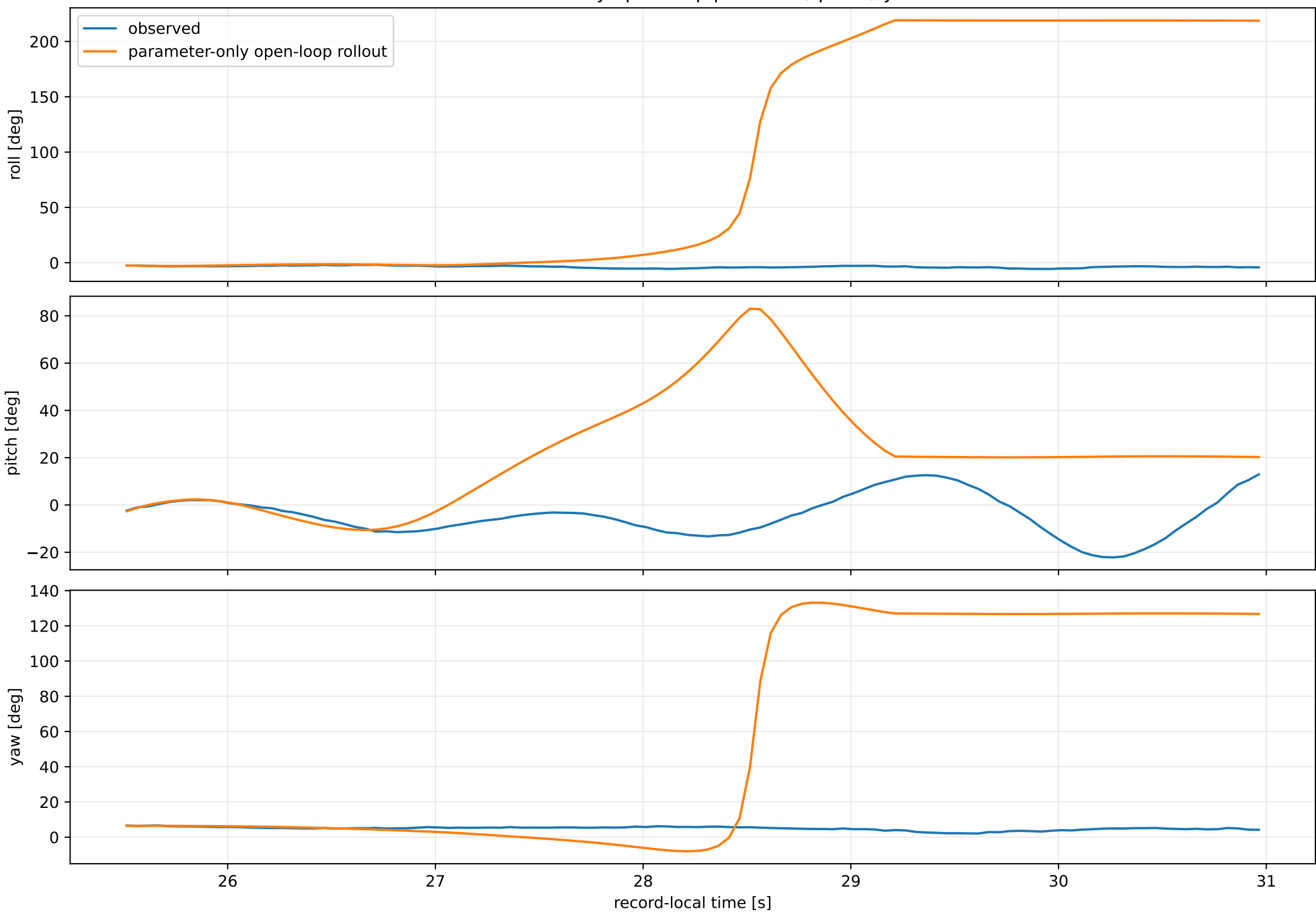
Fine replay error after the 14-D change — lower is better



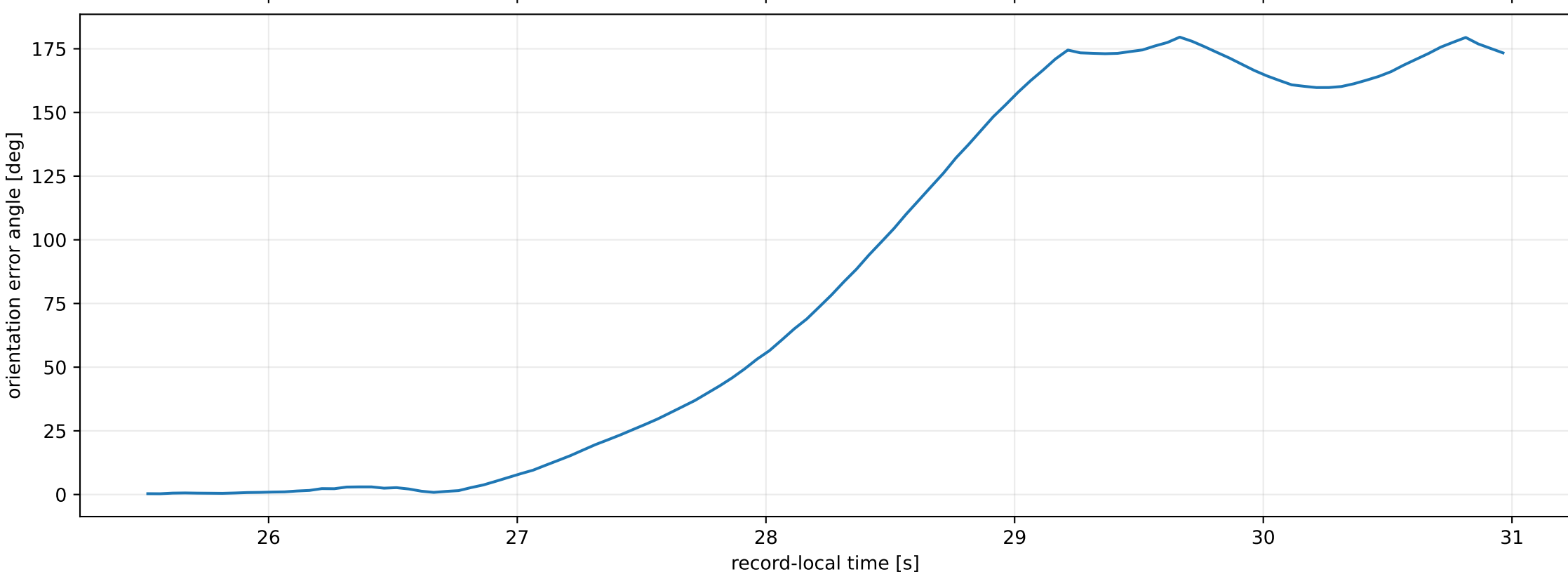
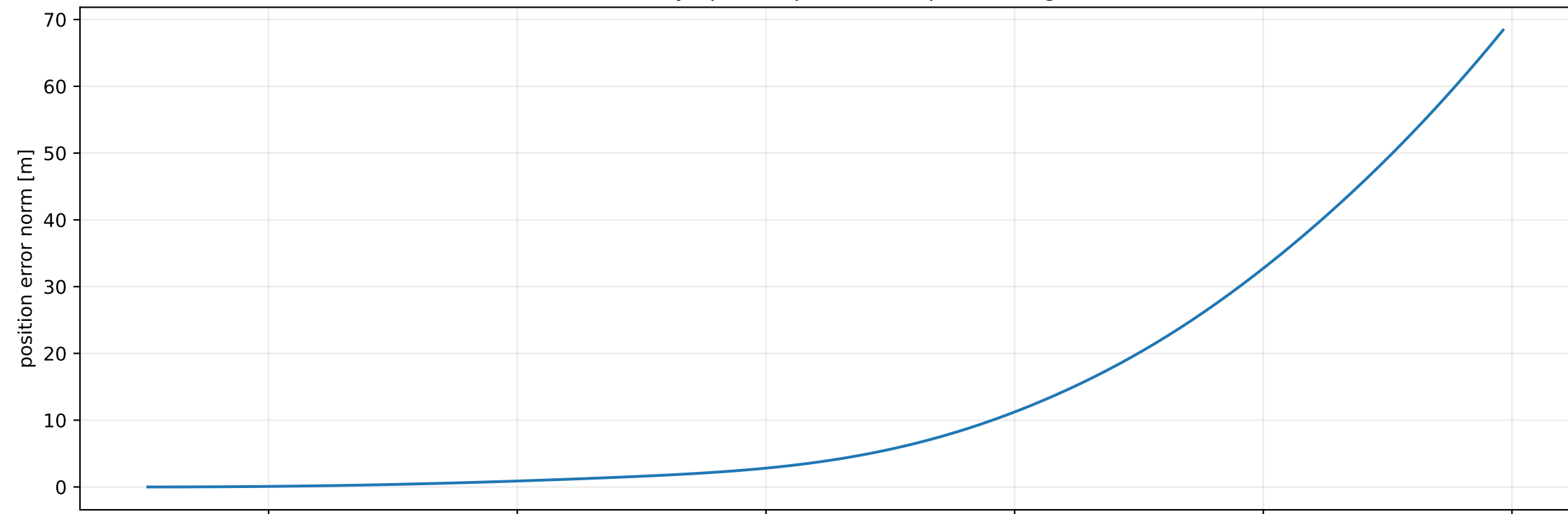
Parameter-only open-loop pose: position



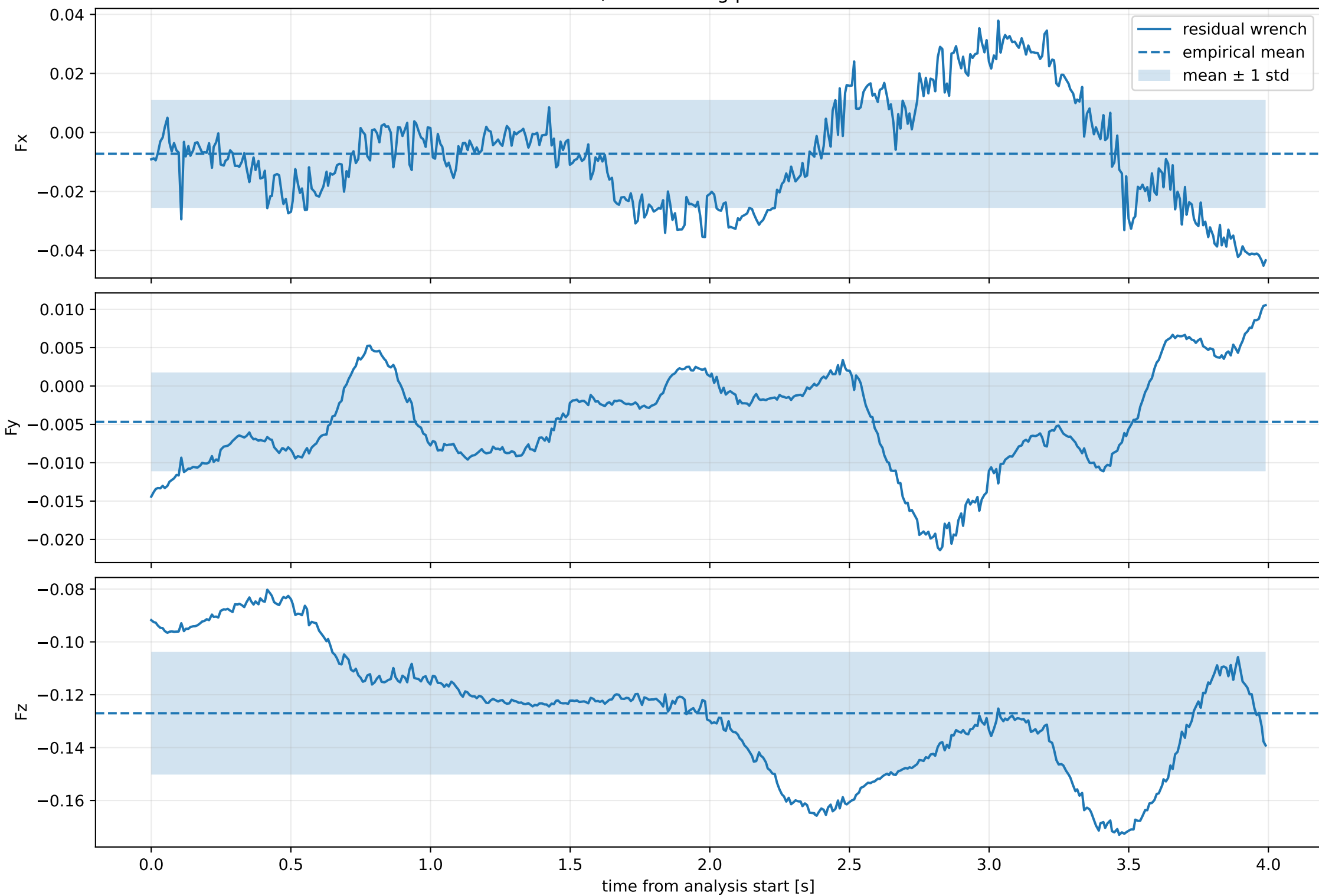
Parameter-only open-loop pose: roll / pitch / yaw



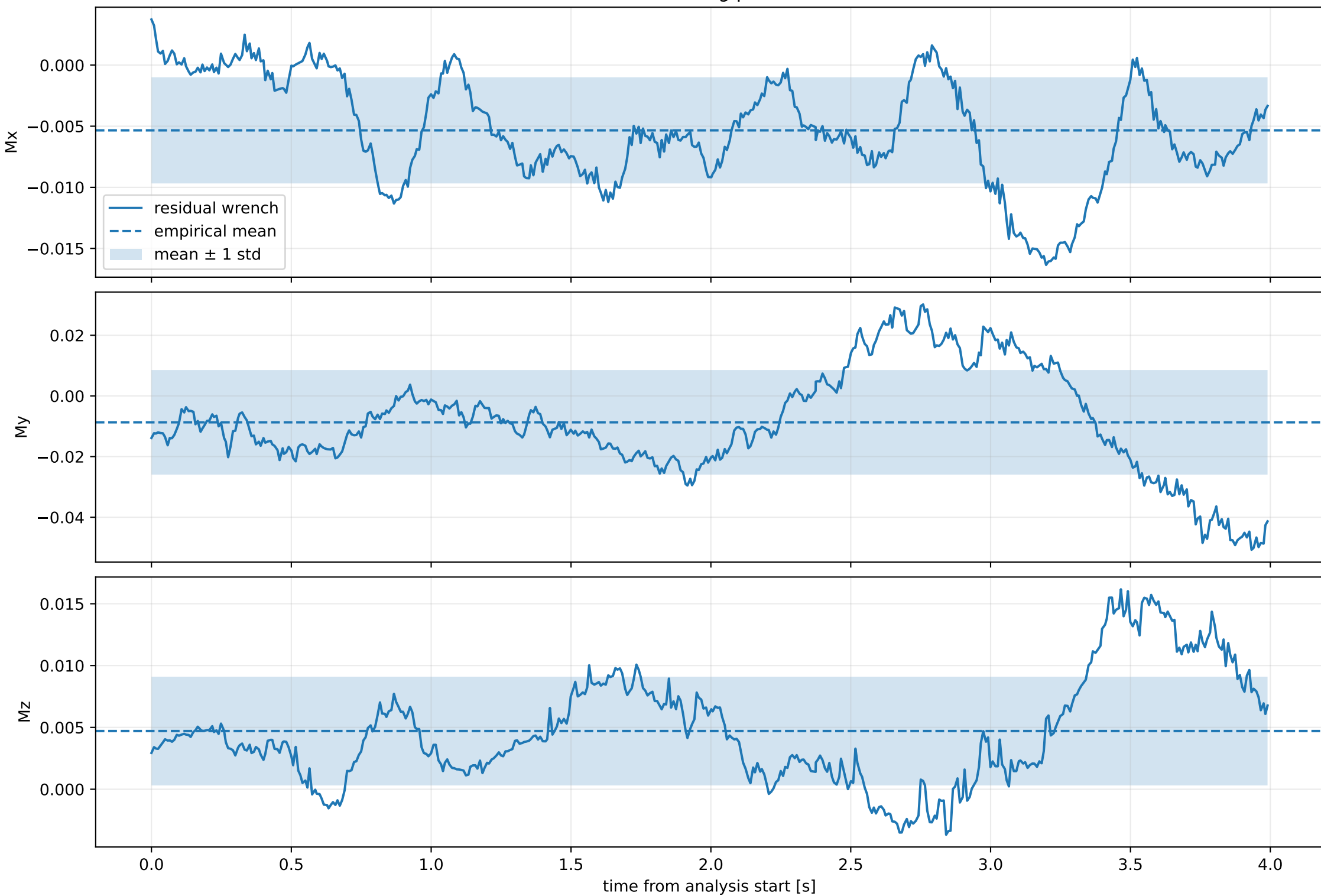
Parameter-only open-loop rollout (separate diagnostic)



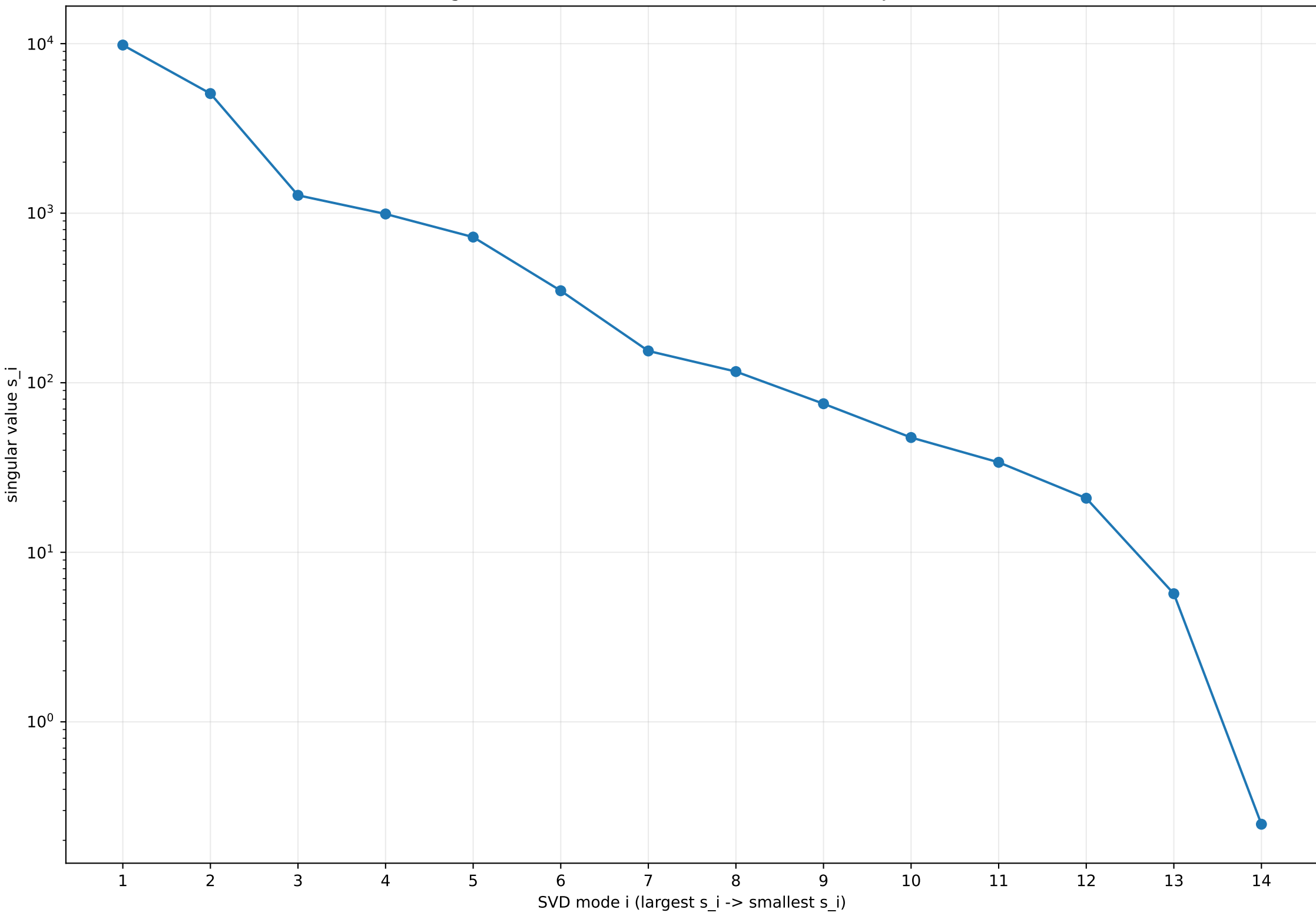
Dimensionless residual force: empirical mean ± 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



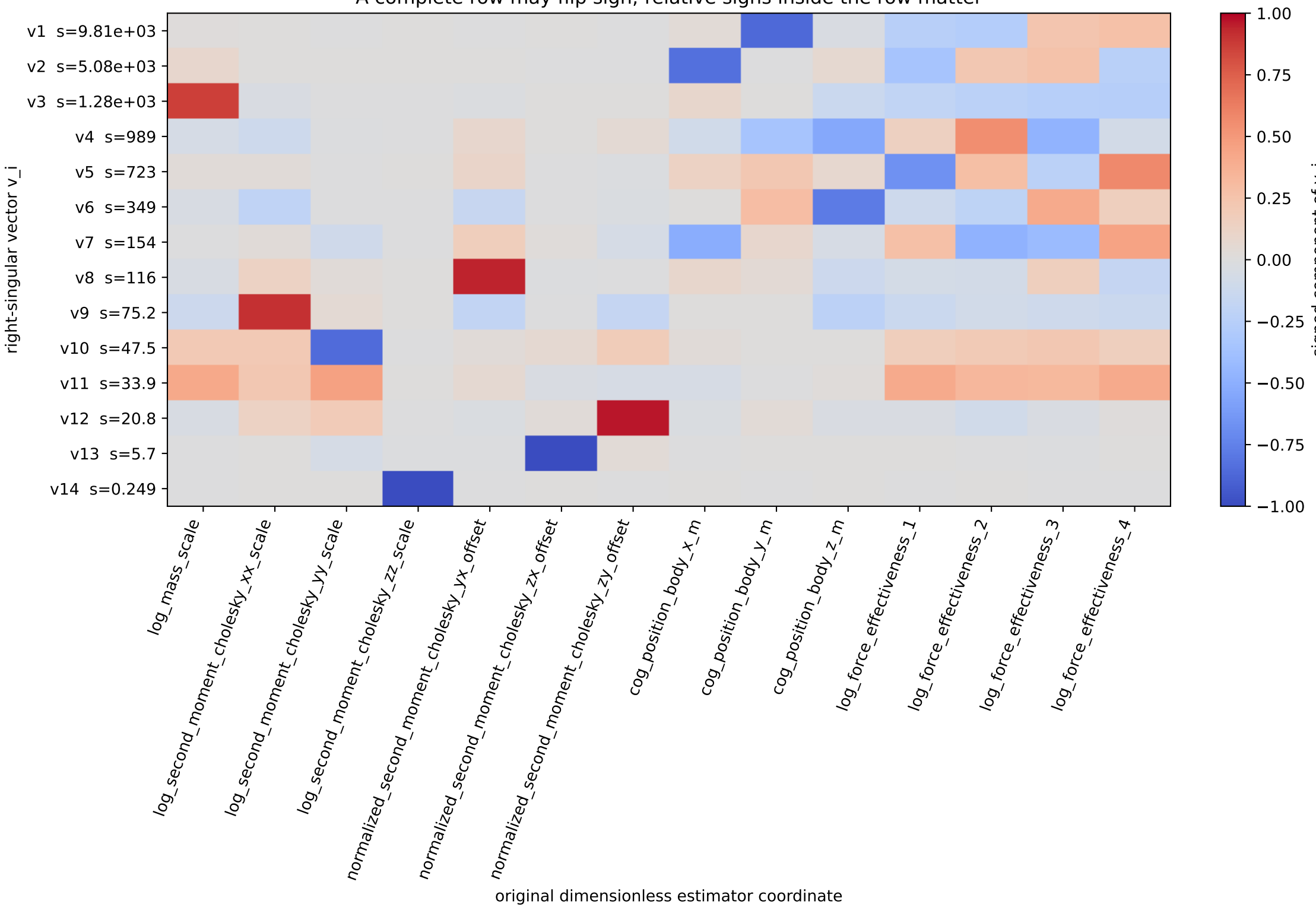
Dimensionless residual torque: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian



Signed right-singular vectors V^T
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.36617815419

inertia J [kg m²]

[0.065002567 3.508271e-06 1.733618e-05]
[3.508271e-06 0.064983027 -3.1119431e-06]
[1.733618e-05 -3.1119431e-06 0.12899253]

principal moments [kg m²]

[0.064982416, 0.065003174, 0.12899254]

CoG position in body frame [m]

[-0.0026364712, 0.00038650867, 0.0097146416]

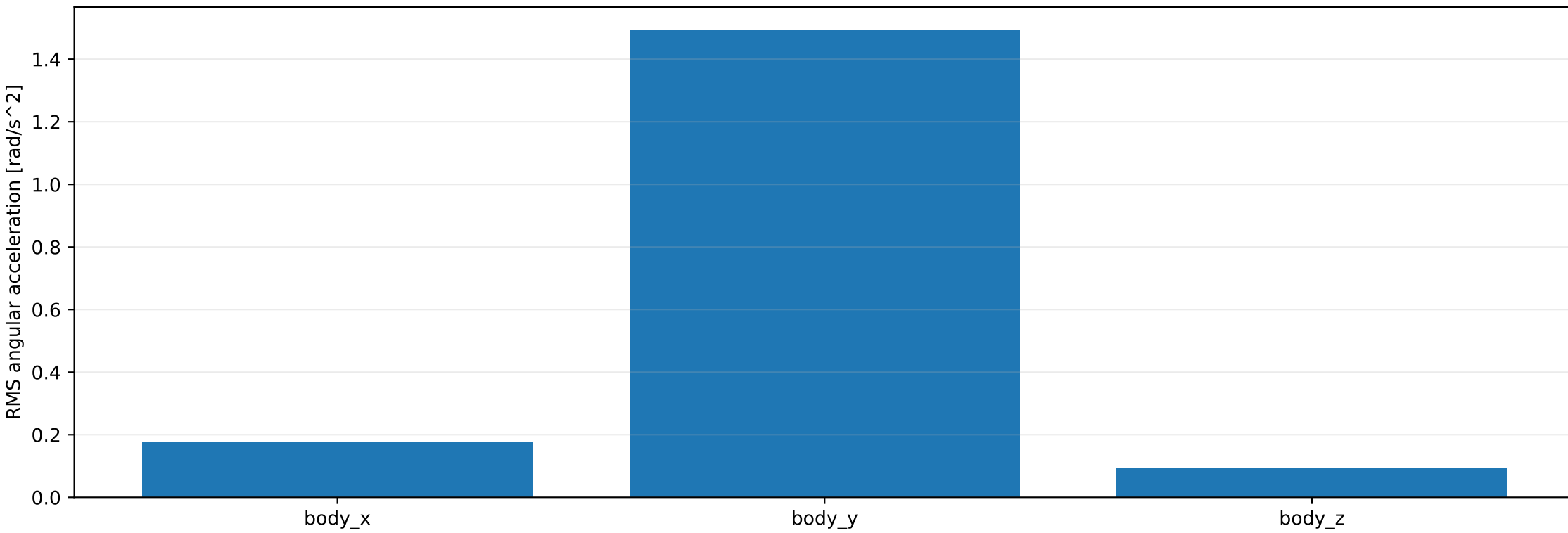
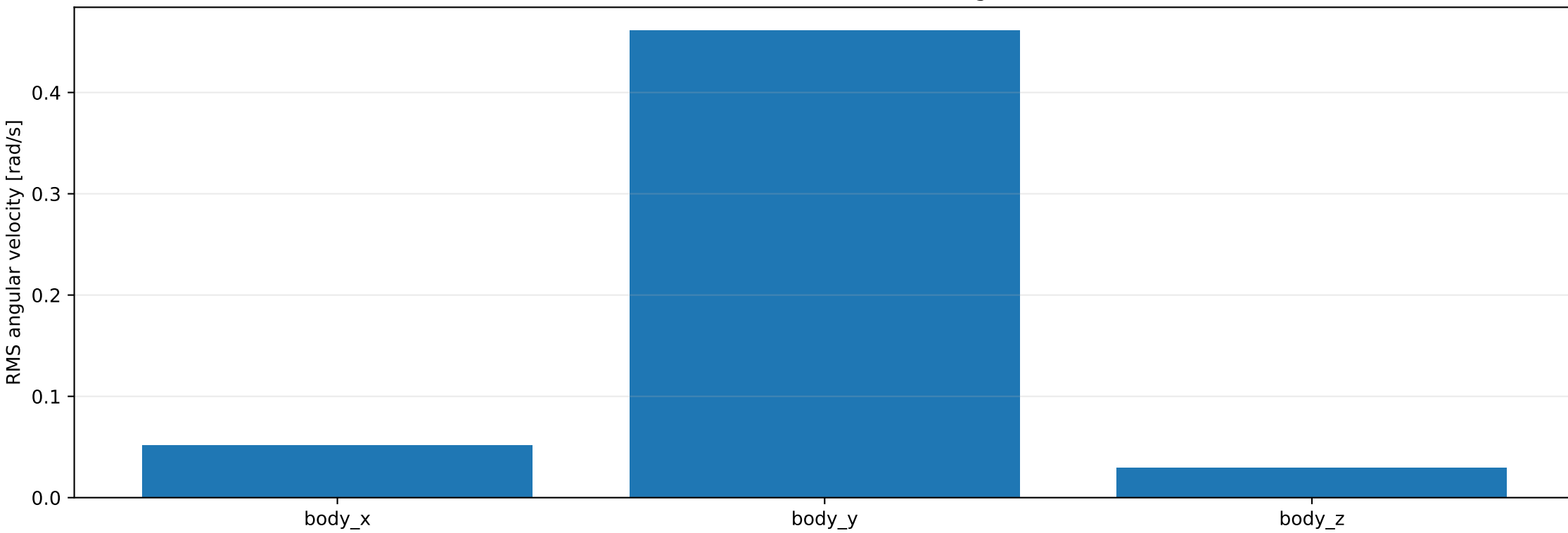
rotor force effectiveness

[0.96227292, 0.97130714, 0.96572508, 0.95650441]

selected delay [s]

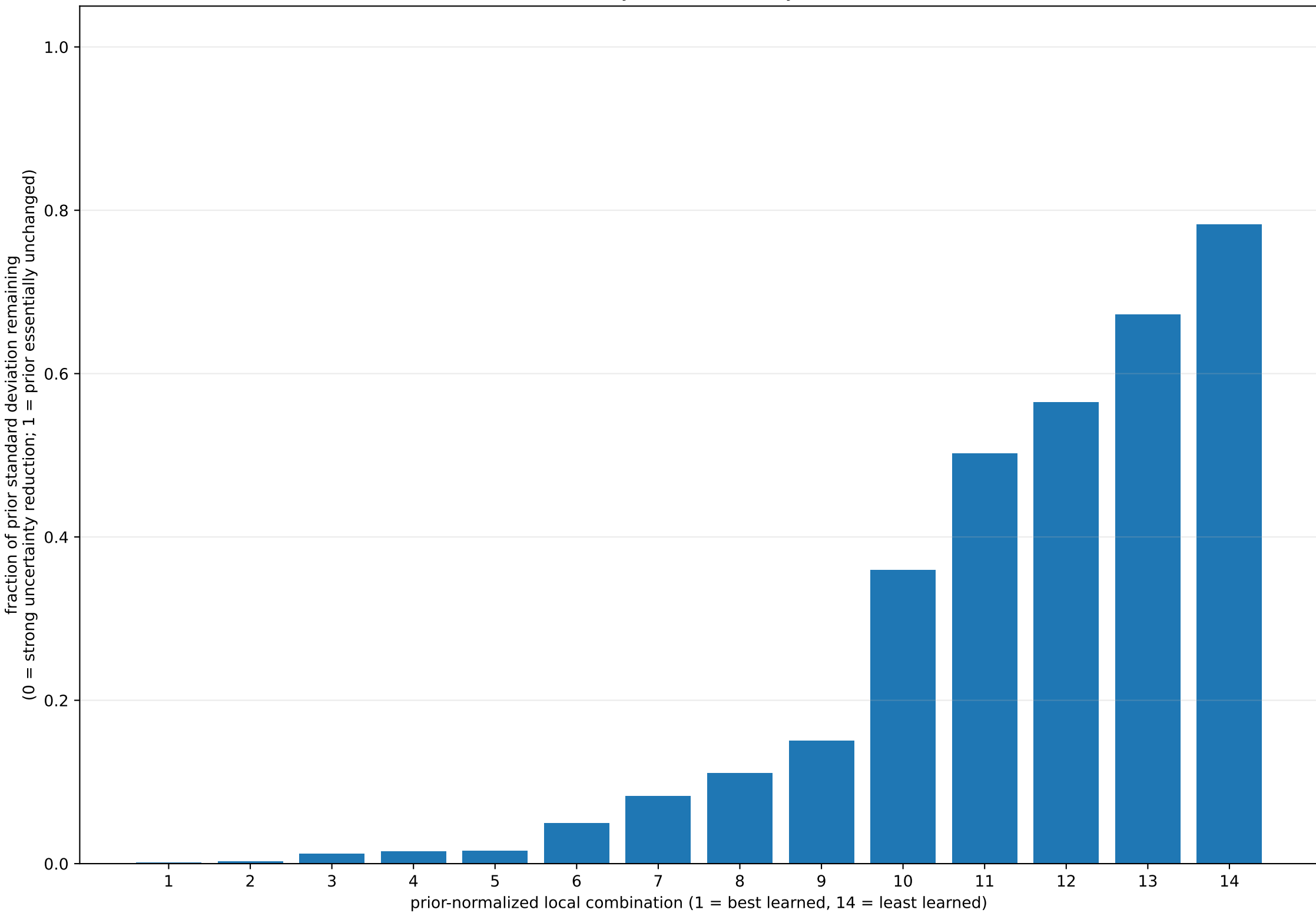
0.2

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

